RAJALAKSHMI ENGINEERING COLLEGE

Dhanusheya G 251801053-2025-AIDS C2

Started on	Thursday, 17 September 2026, 8:27 AM
State	Finished
Completed on	Thursday, 17 September 2026, 8:27 AM
Time taken	37 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

#define MAX_SIZE 100005

int arr1[MAX_SIZE];
int arr2[MAX_SIZE];

void solve() {
    int n1, n2;

    // Read the size of the first array
    if (scanf("%d", &n1) != 1) return;
    // Read the elements of the first array
    for (int i = 0; i < n1; i++) {
        if (scanf("%d", &arr1[i]) != 1) return;
    }

    // Read the size of the second array
    if (scanf("%d", &n2) != 1) return;
    // Read the elements of the second array
    for (int i = 0; i < n2; i++) {
        if (scanf("%d", &arr2[i]) != 1) return;
    }
```

```
24
25      int i = 0, j = 0;
26      int first_print = 1;
27
28      // Two-pointer intersection logic
29      while (i < n1 && j < n2) {
30          if (arr1[i] == arr2[j]) {
31              if (!first_print) {
32                  printf(" ");
33              }
34              printf("%d", arr1[i]);
35              first_print = 0;
36              i++;
37              j++;
38          } else if (arr1[i] < arr2[j]) {
39              i++;
40          } else {
41              j++;
42          }
43      }
44      printf("\n");
45  }
46
47  int main() {
48      int t;
49      if (scanf("%d", &t) != 1) return 0;
50
51      while (t--) {
52          solve();
```

	Input	Expected	Got	
✔	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✔
✔	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course